QBS 103 Final Project - Submission 2

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Submission Notes

Collaborators: none Additional resources used: course materials; google; ggplot2 documentation (https://ggplot2.tidyverse.org); data source: Gonzalez-Vazquez et al. / GSE85307, https://pubmed.ncbi.nlm.nih.gov/27841759/

Choice of genes and covariates

- **Genes:** *LYN* (Affymetrix probe 8146500; an immune-cell signaling kinase), *JAK2* (a kinase in cytokine signaling) and *PTX3* (pentraxin 3, an inflammatory response protein). I chose genes related to immune and inflammatory processes because preeclampsia is associated with maternal immune and inflammatory changes.
- **Continuous covariate:** maternal vitamin D (25-hydroxyvitamin D, ng/mL), `mat_vitD`.
- **Categorical covariate 1:** pregnancy outcome (Normal vs. Preeclampsia), `preg_preeclampsia`.
- **Categorical covariate 2 (derived from a continuous variable):** maternal obesity status (BMI $<$ 30 vs. BMI $>=$ 30 kg/mˆ2), created from `mat_BMI`.

Load and wrangle the data

All wrangling is done in R, starting from the original, unmodified files provided on Canvas (the three .csv files must be in the same folder as this .Rmd).

```
library(ggplot2)

# ---- Read the three original files as provided ----
covar <- read.csv("GSE85307_covar.csv", check.names = FALSE, stringsAsFactors = FALSE)
expr  <- read.csv("GSE85307_geneExpression.csv", check.names = FALSE, stringsAsFactors = FALSE)
genes_link <- read.csv("GSE85307_geneNames.csv", colClasses = "character")

# The first column of the covariate file carries a hidden byte-order mark; rename it cleanly
names(covar)[1] <- "subjectID"

# In the expression file the first (unnamed) column holds the probe IDs: use it as row names
rownames(expr) <- expr[[1]]
expr <- expr[, -1]

# ---- Clean the covariate values ----
# Every covariate value is stored as text such as "age_years: 31.06"; keep only the part after ": "
```

```r
for (v in names(covar)[3:ncol(covar)]) {
  covar[[v]] <- sub("^[^:]*: ", "", covar[[v]])
}

# Convert the continuous covariates to numbers
num_vars <- c("mat_age_yrs", "delivery_gestAge_wks", "sampleCollection_gestAge_wks",
              "mat_BMI", "mat_vitD")
for (v in num_vars) {
  covar[[v]] <- as.numeric(covar[[v]])
}

# ---- Match the covariates to the expression columns using geoID ----
# Put the covariate rows in the same order as the columns of the expression matrix
covar <- covar[match(colnames(expr), covar$geoID), ]

# Verify the matching: no sample is missing and every row lines up with its expression column
stopifnot(!any(is.na(covar$geoID)), all(covar$geoID == colnames(expr)))
cat("Samples matched:", nrow(covar), "of", ncol(expr), "expression columns\n")
```

```
## Samples matched: 157 of 157 expression columns
```

```r
# ---- Build the analysis data frame (one row per sample, one column per gene) ----
gene_names <- c("LYN", "JAK2", "PTX3")

dat <- data.frame(
  geoID   = covar$geoID,
  vitD    = covar$mat_vitD,                       # continuous covariate
  outcome = factor(covar$preg_preeclampsia, levels = c("Normal", "Preeclampsia")),
  BMI     = covar$mat_BMI
)

# Add the expression of each selected gene as its own column, using the probe-to-gene linker
for (g in gene_names) {
  probe <- genes_link$PROBEID[genes_link$GENE == g]
  # exactly one probe per gene, and it must be present in the expression data
  stopifnot(length(probe) == 1, probe %in% rownames(expr))
  dat[[g]] <- as.numeric(expr[probe, ])
}

# Derived categorical variable: obesity status from BMI (obese is BMI >= 30 kg/m^2)
dat$obesity <- factor(ifelse(dat$BMI >= 30, "BMI >= 30", "BMI < 30"),
                      levels = c("BMI < 30", "BMI >= 30"))

# Quick checks without printing the whole data frame
dim(dat)
```

```
## [1] 157   8
```

```r
colSums(is.na(dat))                    # no missing values in the variables used
```

```
##   geoID    vitD outcome     BMI     LYN    JAK2    PTX3 obesity
##       0       0       0       0       0       0       0       0
```

```r
table(dat$outcome, dat$obesity)        # sample size in each boxplot group
```

```
##
##                BMI < 30 BMI >= 30
##   Normal             79        31
##   Preeclampsia       21        26
```

The plotting function

```r
# One shared, clean theme for all figures (white background, no grid, readable text)
theme_pub <- theme_classic(base_size = 13) +
  theme(plot.title    = element_text(face = "bold", size = 14),
        plot.subtitle = element_text(size = 11),
        axis.title    = element_text(face = "bold"),
        axis.text     = element_text(colour = "black"),
        legend.title  = element_text(face = "bold"),
        legend.position = "right")
```

The function `make_gene_plots()` takes (1) a data frame, (2) a vector of one or more gene names (column names of the data frame), (3) the name of one continuous covariate and (4) the names of two categorical covariates. For every gene it returns a histogram, a scatterplot against the continuous covariate, and a boxplot split by both categorical covariates.

```r
make_gene_plots <- function(data, genes, cont_var, cat_vars, labels = NULL) {
  # data     : data frame with one row per sample
  # genes    : character vector of gene names (columns holding the expression values)
  # cont_var : name of the continuous covariate (used on the x-axis of the scatterplot)
  # cat_vars : names of 2 categorical covariates (1st = x-axis, 2nd = fill color of the boxplot)
  # labels   : optional named character vector giving nicer axis labels for the covariates
  # Returns a list with one element per gene; each element holds the three ggplot objects.

  # ---- Check the inputs ----
  stopifnot(is.data.frame(data),
            is.character(genes), length(genes) >= 1,
            all(genes %in% names(data)),
            length(cont_var) == 1, cont_var %in% names(data), is.numeric(data[[cont_var]]),
            length(cat_vars) == 2, all(cat_vars %in% names(data)))

  # Helper: pretty label for a covariate (falls back to the variable name)
  get_label <- function(x) {
    if (!is.null(labels) && x %in% names(labels)) {
      labels[[x]]
    } else {
      x
    }
  }

  # Colorblind-friendly colors for the levels of the 2nd categorical covariate
  fill_levels <- levels(factor(data[[cat_vars[2]]]))
  stopifnot(length(fill_levels) <= 4)
```

```r
  fill_cols <- setNames(c("#56B4E9", "#D55E00", "#009E73", "#E69F00")[seq_along(fill_levels)],
                        fill_levels)

  plots <- list()

  # ---- Make the three plots for each gene ----
  for (g in genes) {
    # Keep only the samples that have all the variables needed for this gene
    d <- data[complete.cases(data[, c(g, cont_var, cat_vars)]), ]
    d[[cat_vars[1]]] <- factor(d[[cat_vars[1]]])
    d[[cat_vars[2]]] <- factor(d[[cat_vars[2]]])
    y_lab <- paste0(g, " expression (log2 intensity)")

    # 1. Histogram of gene expression with a dashed line at the mean
    p_hist <- ggplot(d, aes(x = .data[[g]])) +
      geom_histogram(bins = 20, fill = "#0072B2", colour = "white") +
      geom_vline(xintercept = mean(d[[g]]), linetype = "dashed", linewidth = 0.8) +
      labs(title = paste("Distribution of", g, "expression"),
           subtitle = sprintf("n = %d, mean = %.2f (dashed line)", nrow(d), mean(d[[g]])),
           x = y_lab, y = "Number of samples") +
      theme_pub

    # 2. Scatterplot of expression vs. the continuous covariate.
    #    The correlation is written in the subtitle (above the panel) so that it can
    #    never overlap with any of the points.
    ct <- cor.test(d[[cont_var]], d[[g]])
    p_txt <- ifelse(ct$p.value < 0.001, "< 0.001", sprintf("= %.3f", ct$p.value))
    p_scatter <- ggplot(d, aes(x = .data[[cont_var]], y = .data[[g]])) +
      geom_point(colour = "#0072B2", alpha = 0.7, size = 2) +
      geom_smooth(method = "lm", formula = y ~ x, colour = "black", fill = "grey70") +
      labs(title = paste(g, "expression vs.", get_label(cont_var)),
           subtitle = sprintf("Pearson r = %.2f, p %s", ct$estimate, p_txt),
           x = get_label(cont_var), y = y_lab) +
      theme_pub

    # 3. Boxplot of expression by both categorical covariates, with the samples overlaid
    p_box <- ggplot(d, aes(x = .data[[cat_vars[1]]], y = .data[[g]], fill = .data[[cat_vars[2]]])) +
      geom_boxplot(outlier.shape = NA, alpha = 0.8, width = 0.7,
                   position = position_dodge(width = 0.8)) +
      geom_point(aes(group = .data[[cat_vars[2]]]),
                 position = position_jitterdodge(jitter.width = 0.15, dodge.width = 0.8),
                 size = 1.4, alpha = 0.6) +
      scale_fill_manual(values = fill_cols, name = get_label(cat_vars[2])) +
      labs(title = paste(g, "expression by", tolower(get_label(cat_vars[1])), "and",
                         tolower(get_label(cat_vars[2]))),
           x = get_label(cat_vars[1]), y = y_lab) +
      theme_pub

    plots[[g]] <- list(histogram = p_hist, scatter = p_scatter, boxplot = p_box)
  }

  return(plots)
}
```

```r
# Nicer axis labels for the covariates (only used for the text on the plots)
cov_labels <- c(vitD = "Maternal vitamin D (ng/mL)",
                outcome = "Pregnancy outcome",
                obesity = "Obesity status")

# Call the function once for all three genes
all_plots <- make_gene_plots(data = dat,
                             genes = gene_names,
                             cont_var = "vitD",
                             cat_vars = c("outcome", "obesity"),
                             labels = cov_labels)
```

Histograms of gene expression

```r
# Show the histogram of each gene (the same loop works for any number of genes)
for (g in gene_names) {
  print(all_plots[[g]]$histogram)
}
```

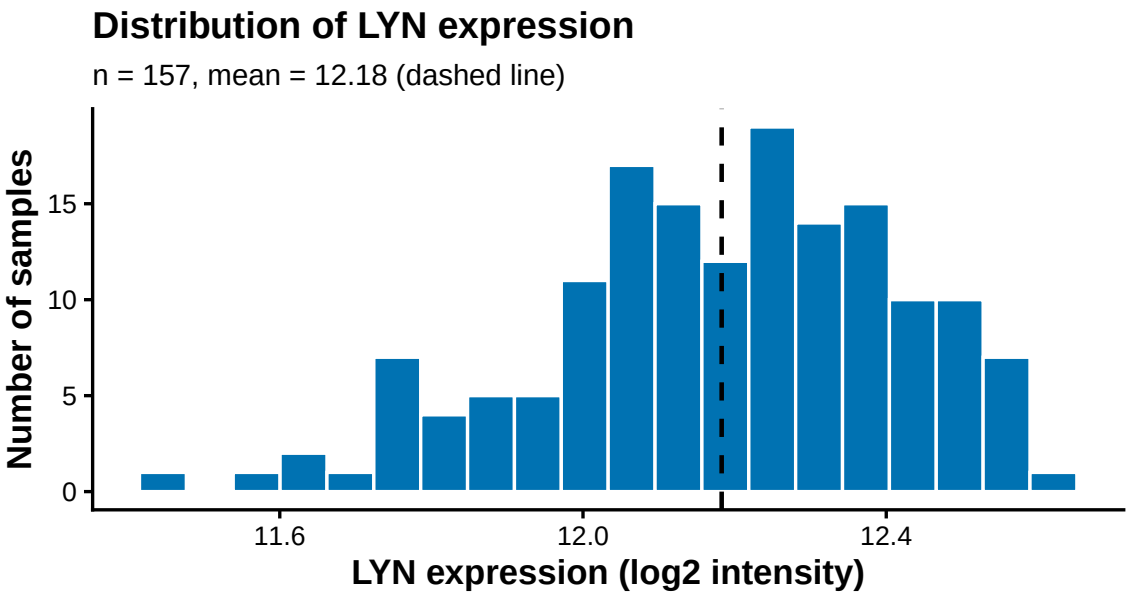


Figure 1: Distribution of LYN expression (log2 intensity) in whole-blood samples from 157 pregnant women. The dashed line marks the mean.

Gene expression vs. maternal vitamin D

```r
for (g in gene_names) {
  print(all_plots[[g]]$scatter)
}
```

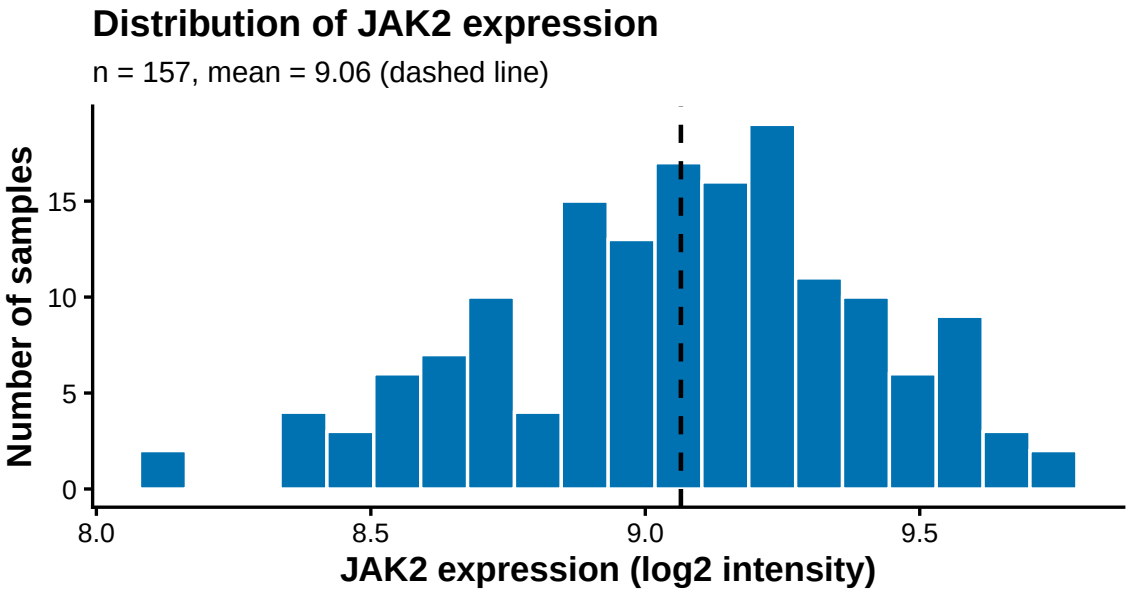


Figure 2: Distribution of JAK2 expression (log2 intensity) in whole-blood samples from 157 pregnant women. The dashed line marks the mean.

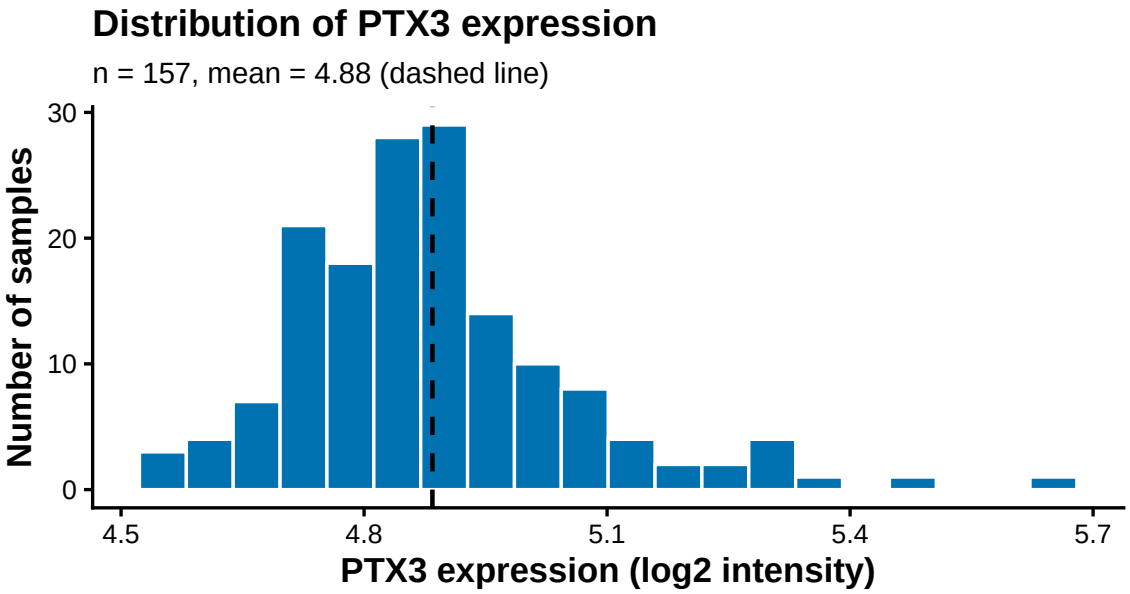


Figure 3: Distribution of PTX3 expression (log2 intensity) in whole-blood samples from 157 pregnant women. The dashed line marks the mean.

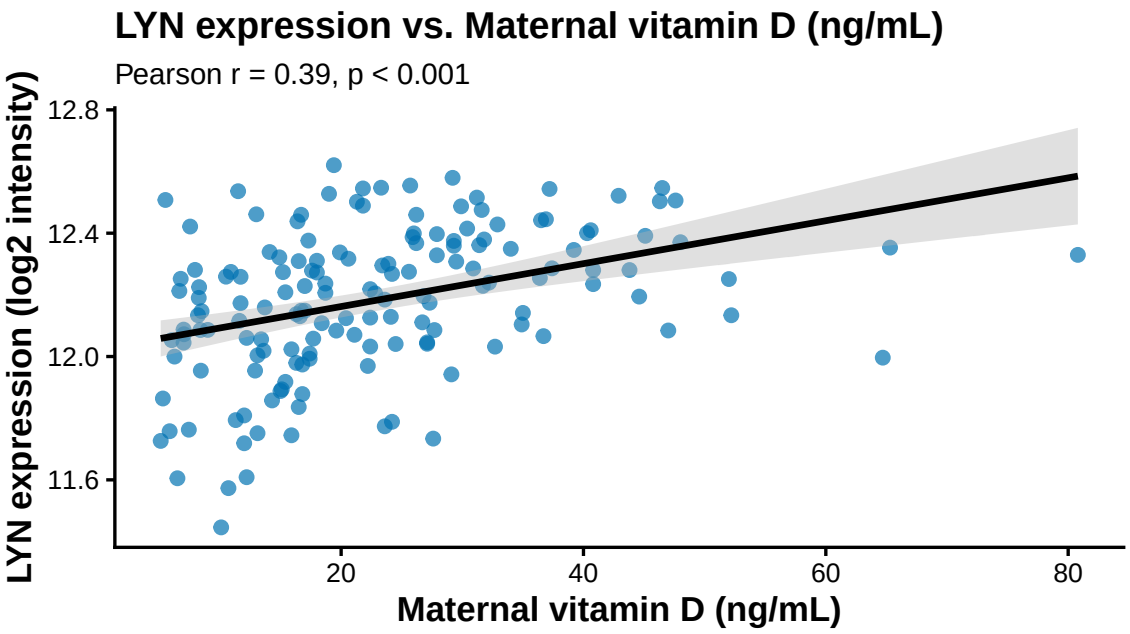


Figure 4: Association between maternal vitamin D (ng/mL) and LYN expression (log2 intensity). Each point is one woman; the line is the linear fit with its 95% confidence band; the Pearson correlation is given above the plot.

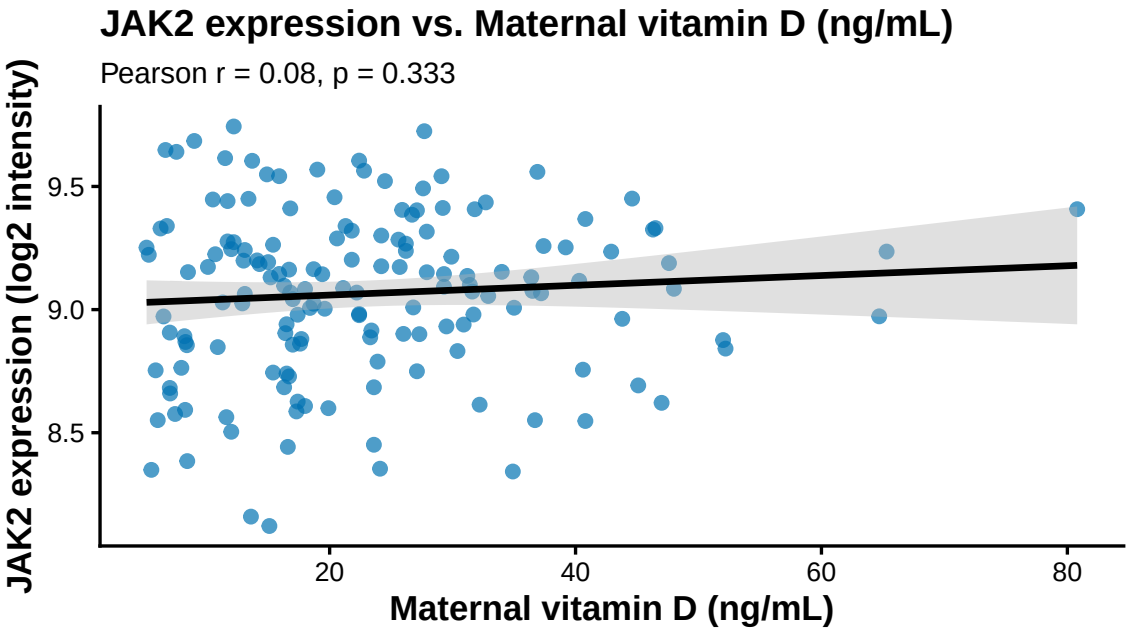


Figure 5: Association between maternal vitamin D (ng/mL) and JAK2 expression (log2 intensity). Each point is one woman; the line is the linear fit with its 95% confidence band; the Pearson correlation is given above the plot.

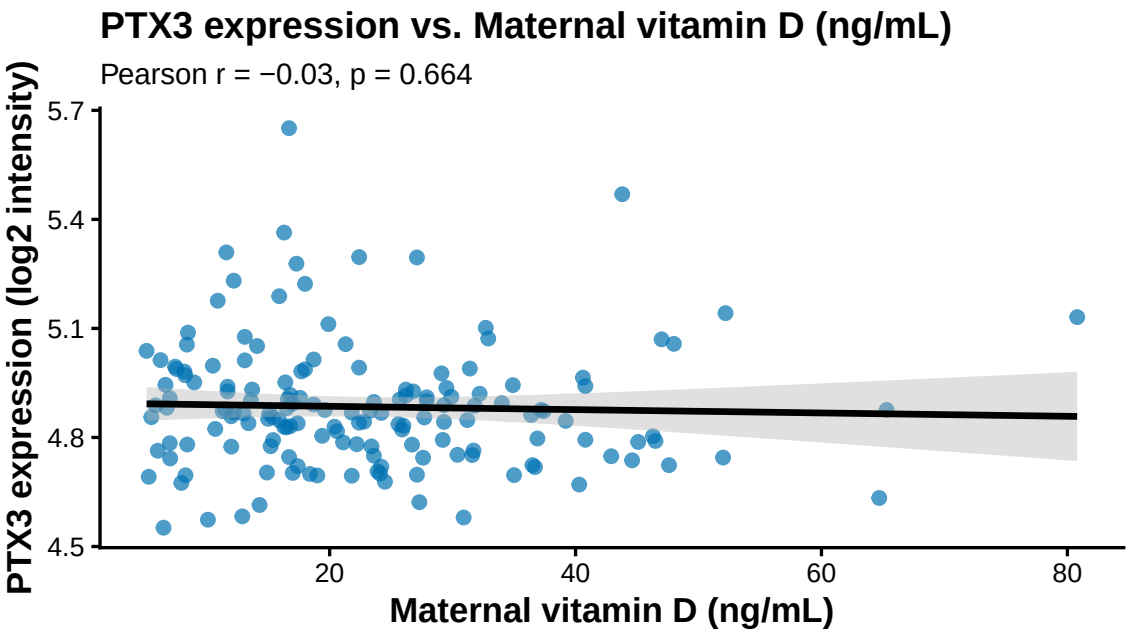


Figure 6: Association between maternal vitamin D (ng/mL) and PTX3 expression (log2 intensity). Each point is one woman; the line is the linear fit with its 95% confidence band; the Pearson correlation is given above the plot.

Gene expression by pregnancy outcome and obesity status

```
for (g in gene_names) {
  print(all_plots[[g]]$boxplot)
}
```

Comparison across the three genes

```
# One summary row per gene: distribution of expression and association with vitamin D
comp <- data.frame(
  Gene = gene_names,
  Mean = sapply(gene_names, function(g) mean(dat[[g]])),
  SD   = sapply(gene_names, function(g) sd(dat[[g]])),
  r_vitD = sapply(gene_names, function(g) cor(dat$vitD, dat[[g]])),
  p_vitD = sapply(gene_names, function(g) cor.test(dat$vitD, dat[[g]])$p.value),
  row.names = NULL
)
comp$p_vitD <- ifelse(comp$p_vitD < 0.001, "< 0.001", sprintf("%.3f", comp$p_vitD))
knitr::kable(comp, digits = 2,
             col.names = c("Gene", "Mean", "SD", "r (vitamin D)", "p"),
             caption = "Summary of the three genes: expression (log2 intensity) and correlation with ma
```

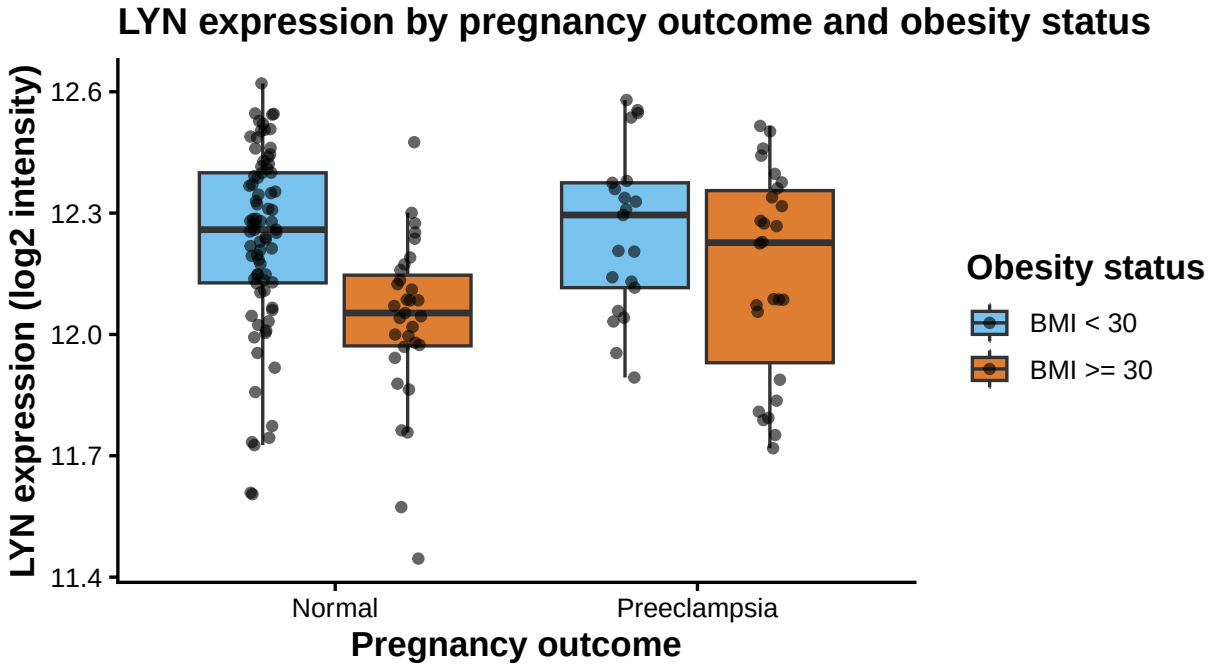


Figure 7: LYN expression (log2 intensity) by pregnancy outcome (Normal vs. Preeclampsia) and maternal obesity status (BMI $<$ 30 vs. $>=$ 30 kg/m^2). Boxes show the median and interquartile range, whiskers extend to 1.5 times the interquartile range, and points are individual samples.

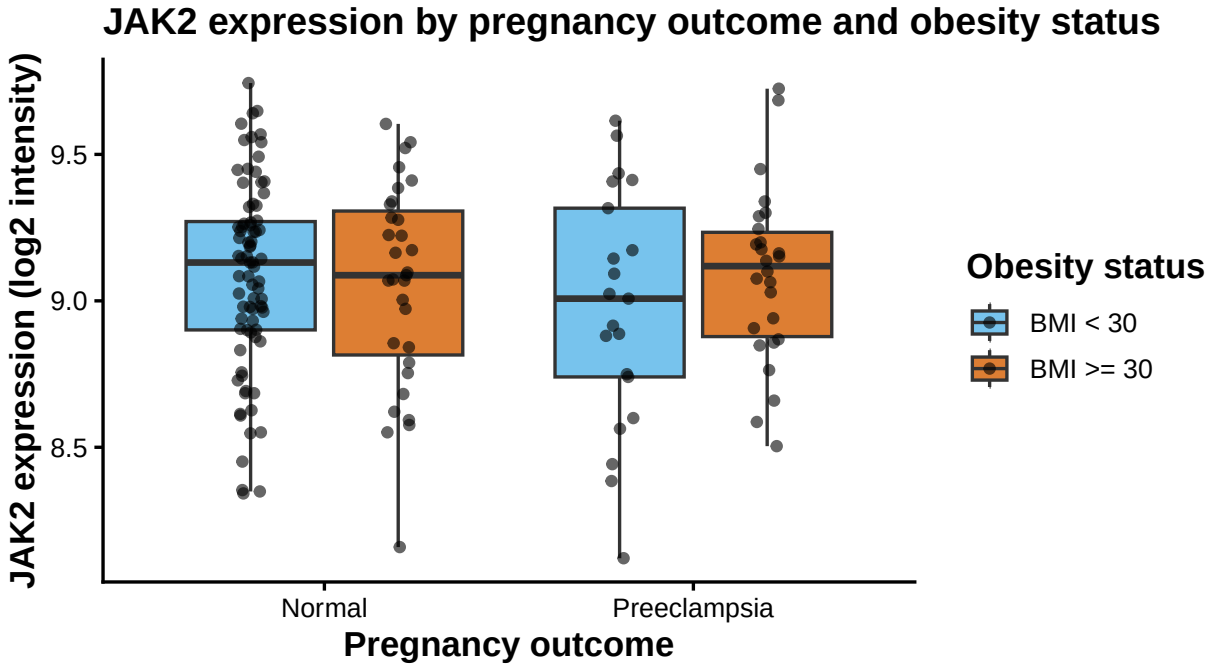


Figure 8: JAK2 expression (log2 intensity) by pregnancy outcome (Normal vs. Preeclampsia) and maternal obesity status (BMI $<$ 30 vs. $>=$ 30 kg/m^2). Boxes show the median and interquartile range, whiskers extend to 1.5 times the interquartile range, and points are individual samples.

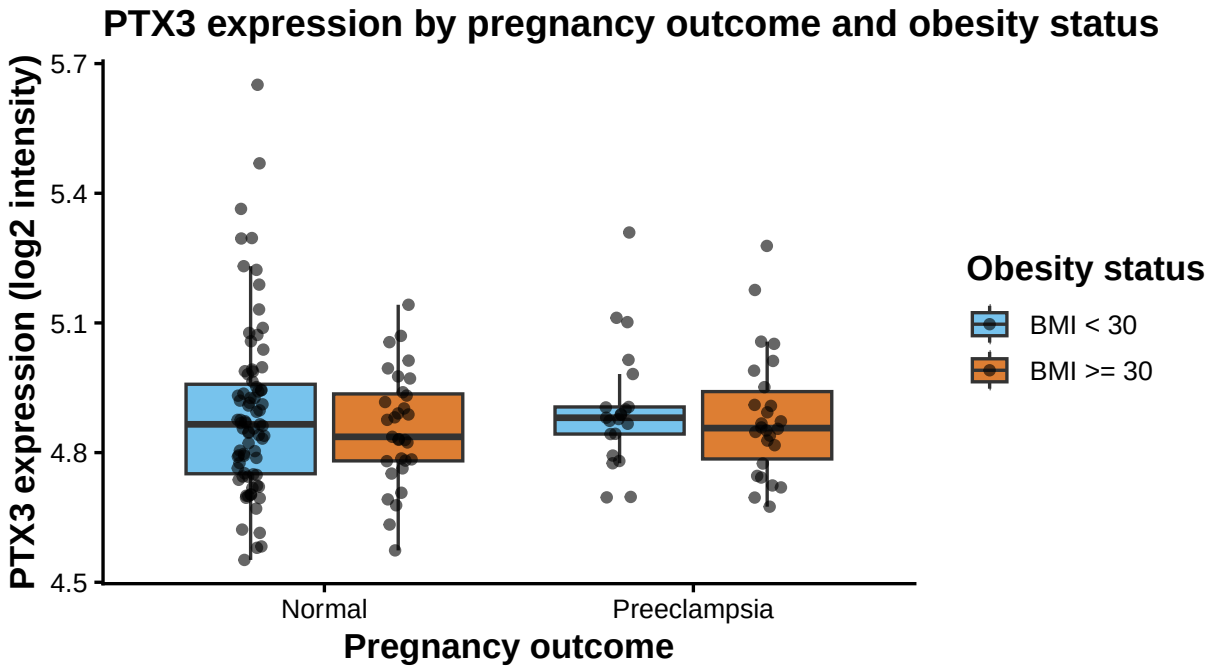


Figure 9: PTX3 expression (log2 intensity) by pregnancy outcome (Normal vs. Preeclampsia) and maternal obesity status (BMI $<$ 30 vs. $>=$ 30 kg/m^2). Boxes show the median and interquartile range, whiskers extend to 1.5 times the interquartile range, and points are individual samples.

Table 1: Summary of the three genes: expression (log2 intensity) and correlation with maternal vitamin D.

Gene	Mean	SD	r (vitamin D)	p
LYN	12.18	0.24	0.39	$<$ 0.001
JAK2	9.06	0.34	0.08	0.333
PTX3	4.88	0.17	-0.03	0.664

Across the three genes, the histograms show how the expression of each gene is spread (Figures 1-3), the scatterplots show whether it moves with vitamin D (Figures 4-6; Pearson r = 0.39, 0.08, -0.03 for LYN, JAK2, PTX3), and the boxplots show how it differs between the obesity groups within each pregnancy outcome (Figures 7-9). The median difference in expression (BMI $>=$ 30 minus BMI $<$ 30, in the whole sample) is -0.18, 0.01, -0.02 for LYN, JAK2, PTX3, respectively. These are descriptive comparisons only. Note that each gene has its own y-axis range, so the heights of the plots should not be compared directly between genes.